Automotive Sales Analytics Insights Report

Tableau project: sales trends, dealer performance, product demand and customer behaviour

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Tool	Tableau

1. Business context and approach

A multi-dealer automobile company sells cars from many manufacturers across several regions. Management lacked clear visibility into which vehicles earn the most revenue, which regions perform best and which customer segments drive sales. This project turns the sales dataset into three interactive Tableau dashboards and a set of data-supported insights.

Dataset at a glance

Measure	Value	Measure	Value
Total revenue	$671.5M	Companies / models	30 / 154
Cars sold	23,906	Dealers	28
Average car price	$28,090	Regions	7
Period	2 Jan 2022 - 31 Dec 2023	Customers (male / female)	78.6% / 21.4%

Total Revenue	Total Cars Sold	Average Car Price
$671.5M	**23,906**	**$28,090**

Data preparation

The CSV was connected to Tableau and checked: 23,906 rows, no missing values and no duplicate Car_id values. The Engine field contained an encoding error in one value ("Double Overhead Camshaft" with extra characters), which was cleaned in a calculated field so the field has exactly two clean values. Year, Quarter and Month fields were created from the purchase date.

Calculated fields and controls built

Field	Purpose
Revenue, Cars Sold, Average Car Price	SUM of price, count of Car_id, average of price: the three KPIs
Income Group	Low Income below 30,000; Middle Income below 70,000; otherwise High Income (assignment thresholds)
Sale Category	Premium Sale if the sale price is above $30,000, otherwise Standard Sale (a price label, not profit)
Top N Dealers (parameter)	Viewer chooses 3 to 28 dealers; default 10. Used with Dealer Rank and Top N Filter
Model Rank, Top 10 Models Filter	Keeps the Top Models chart at 10 bars even when the Model filter is used
Region, Company, Model, Date filters	Applied to every worksheet, so one filter changes all charts and KPIs

The three dashboards (each 1280 x 800, with the four filters)

Dashboard	What it shows
Executive Sales Overview	Total Revenue, Total Cars Sold and Average Car Price cards; monthly sales trend; revenue by region (clickable as a filter)
Customer Behaviour	Annual income vs sale price scatter; vehicles sold by body style and gender; body style distribution donut
Dealer & Product Performance	Top N dealers by revenue (with slider); top 10 car models; transmission preference

2. Key insights

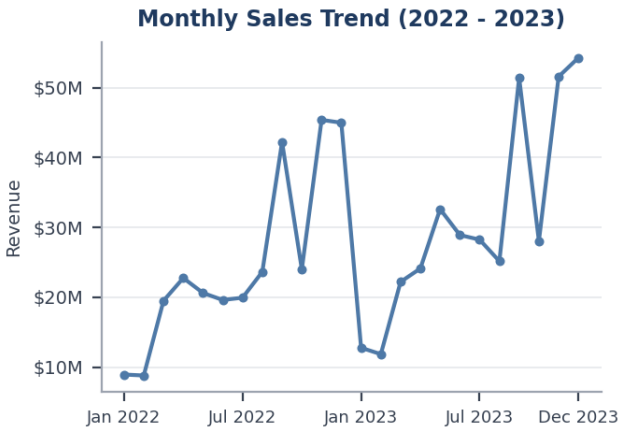
Each insight separates what the data shows (fact) from what management could consider (recommendation). The charts follow the worksheets built in Tableau.

INSIGHT 1

Revenue and volume grew in 2023

Fact. 2023 revenue was about $371.2M against $300.3M in 2022, up 23.6%. Cars sold rose from 10,645 to 13,261 (up 24.6%).

Recommendation. Find out whether the growth came from higher demand or from more dealer capacity before setting 2024 targets.

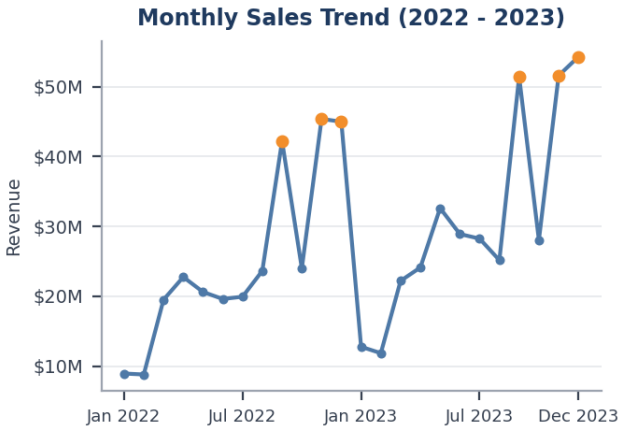


INSIGHT 2

Sales peak in September, November and December

Fact. These three months (orange points) together account for about 43% of all cars sold and 43% of revenue, in both years. The fourth quarter alone is about 37% of revenue.

Recommendation. Plan inventory, staffing and promotions ahead of the September to December period.

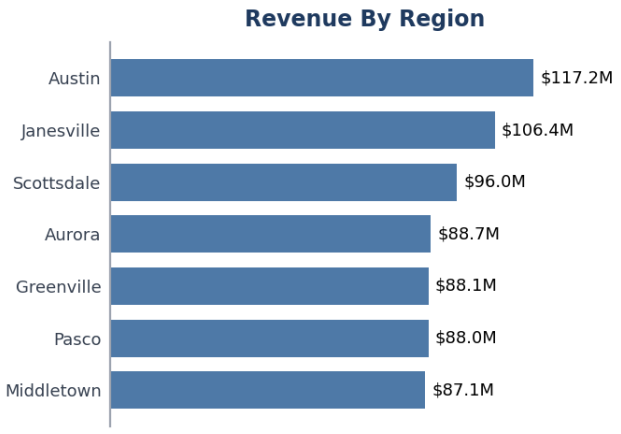


INSIGHT 3

Austin is the strongest region

Fact. Austin earned about $117.2M (17.5% of revenue), about $10.8M more than second-placed Janesville. The four lowest regions are within about $1.6M of each other.

Recommendation. Study what Austin does differently and test whether it works in the weaker regions.



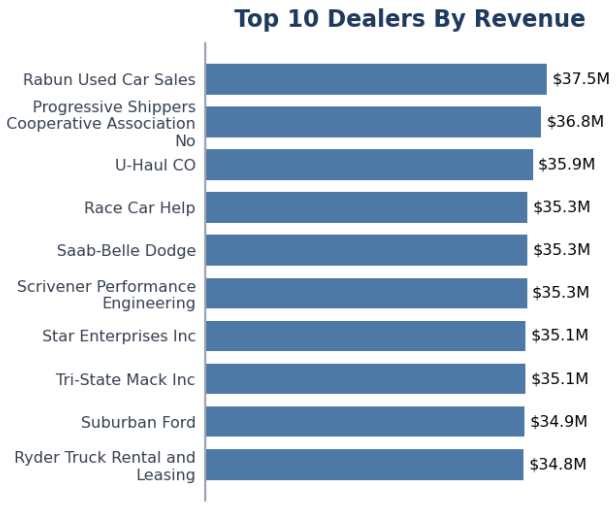
INSIGHT 4

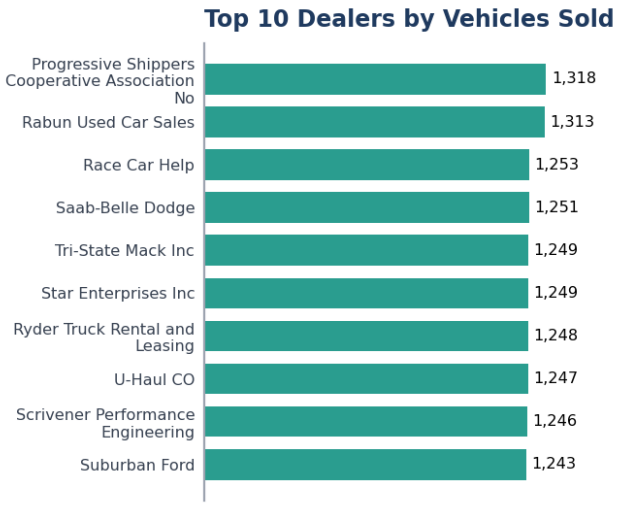
Dealer revenue falls into two tiers

Fact. The top 10 dealers each earn between $34.8M and $37.5M and together make up 53.0% of revenue. The 11th dealer earns $17.9M, about half of the 10th, and the other 18 dealers earn $17.1M to $17.9M each. Rabun Used Car Sales leads in revenue; Progressive Shippers Cooperative Association No leads in cars sold (1,318), so the dealer with the most cars is not the one with the most revenue.

Recommendation. Investigate what the top-tier dealers have in common, such as location, size or stock mix.

Careful. The data cannot say why the gap exists.



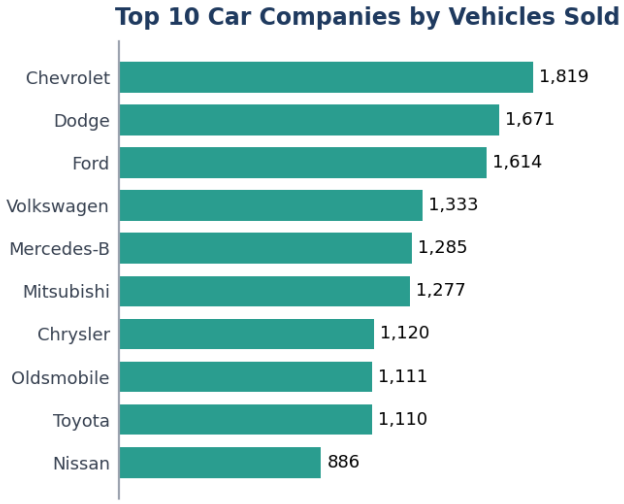


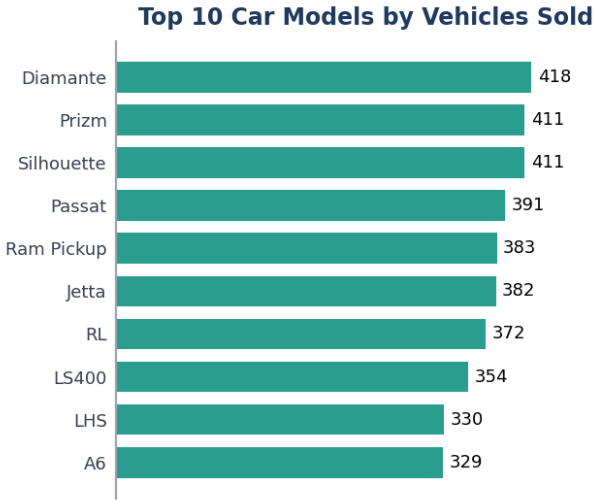
INSIGHT 5

Chevrolet, Dodge and Ford lead, but demand is spread

Fact. Chevrolet sold 1,819 cars, Dodge 1,671 and Ford 1,614, together about 21% of all cars sold, out of 30 manufacturers. Model demand is close too: Diamante leads with 418, then Prizm and Silhouette with 411 each.

Recommendation. Keep stock balanced across the leading brands rather than relying on one manufacturer.



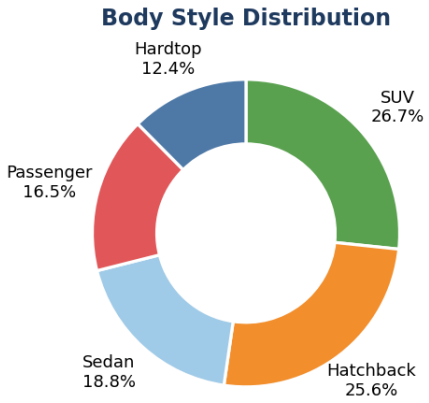


INSIGHT 6

SUV and Hatchback are the most popular body styles

Fact. SUV makes up 26.7% of cars sold and Hatchback 25.6%. Sedan is 18.8%, Passenger 16.5% and Hardtop 12.4%.

Recommendation. Prioritise SUV and Hatchback stock, without cutting the other styles sharply.



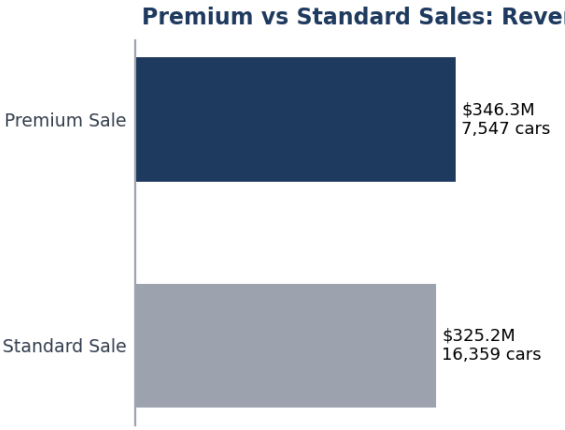
INSIGHT 7

Premium-priced sales carry about half of revenue

Fact. Sales above $30,000 are 31.6% of cars sold (7,547 cars) but 51.6% of revenue (about $346.3M).

Recommendation. Track the premium share over time and look at which models drive it.

Careful. This is a sale-price label. The dataset has no cost data, so it says nothing about profit.



INSIGHT 8

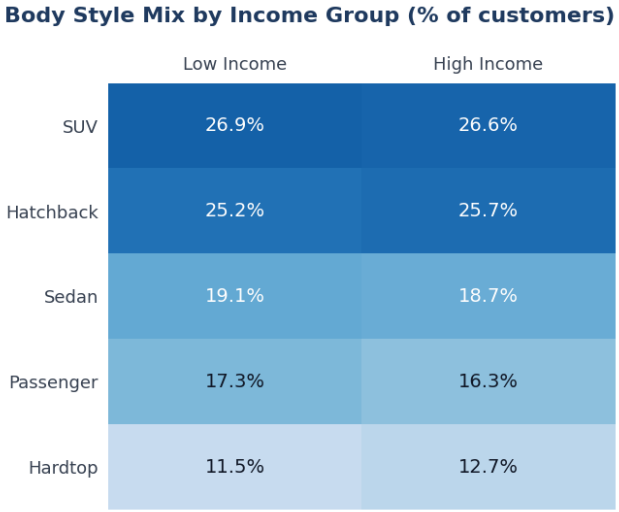
Income shows no link to price in this data

Fact. The correlation between annual income and sale price is about 0.01, and the trend line in the scatter plot is almost flat. Low and High income customers buy body styles in nearly the same proportions (SUV 26.9% and 26.6%, Hatchback 25.2% and 25.7%).

Recommendation. Do not use income alone to target premium cars; test other factors such as model and body style.

Careful. No customers fall in the Middle Income band under the assignment thresholds, so Income Group compares only Low (5,275) and High (18,631). The result describes these sales only.





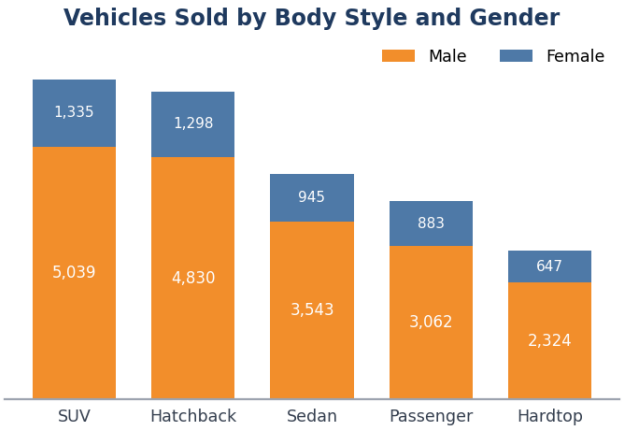
INSIGHT 9

Gender mix is almost the same across body styles

Fact. About 79% of customers are male. The female share stays between 20.9% and 22.4% in every body style.

Recommendation. Gender does not appear to change body style choice here, so campaigns need not be split by gender for body style.

Careful. The data shows the buyers in this dataset, not the wider market.



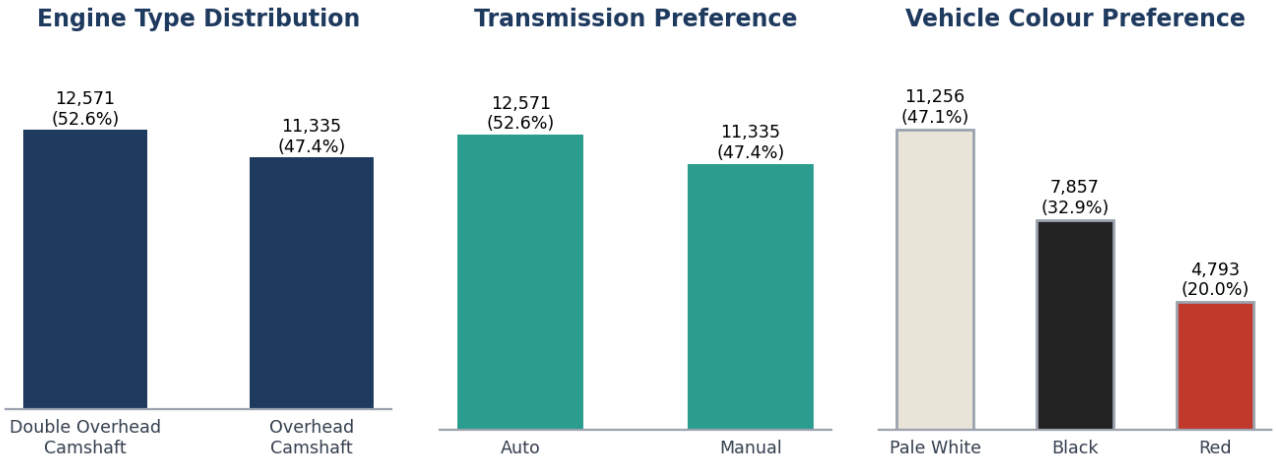
INSIGHT 10

Engine, transmission and colour

Fact. Automatic cars are 52.6% of sales and manual 47.4%. Every Double Overhead Camshaft car is automatic (12,571) and every Overhead Camshaft car is manual (11,335), so the two charts show the same split. Pale White is the top colour at 47.1%, ahead of Black (32.9%) and Red (20.0%).

Recommendation. Plan engine and transmission as one choice, since they move together. Keep Pale White as the lead colour in stock.

Careful. The data holds only three colours, and sales do not show what customers wanted but could not find.



3. Recommendations and limitations

What management could consider

- Prepare inventory and staffing before the September to December peak, which carries about 43% of sales.
- Learn from Austin and from the top 10 dealers, which earn about twice as much as the other 18.
- Lead with SUV and Hatchback stock and keep the leading brands (Chevrolet, Dodge, Ford) well stocked.
- Track the premium sale share (over $30,000), which brings in about half of revenue from under a third of cars.
- Investigate drivers other than income and gender if the goal is to predict who buys expensive cars.

Limitations of the analysis

- The data holds sales only: no costs, profit, stock levels or lost sales, so conclusions are about what was sold, not what customers wanted.
- Sale Category is a price label (over $30,000), not a profitability measure.
- Income Group has no Middle Income customers under the given thresholds, because most incomes are far above 70,000.
- Engine type and transmission match one-to-one, so they cannot be used to explain each other.
- Patterns are associations in this dataset. They do not prove cause and effect.